

Systems that refuse to invent.

Every system I use will guess, then deliver the guess in the same voice it uses for facts. That is not a bug. **An honest “I don't know” is harder to build than a confident answer,** and nobody is rewarded for building it.

So that is what I build. Five projects, one idea, arrived at five separate times before I noticed it had become a thesis.

The last page is the rest of me: two years of running events, teaching, and shipping code to people who owed me nothing.



HARVARD COLLEGE PROJECT FOR ASIAN AND INTERNATIONAL RELATIONS

Office of the President & Executive Board
39 Shaght Street, Cambridge, MA 02138
United States of America

Dear Kunal Kumar:

Congratulations! The admissions team at the Harvard College Project for Asian and International Relations (HCAIR) is overjoyed to let you know that we have selected you as a delegate for the 2026 HCAIR Asia Conference!

The team has diligently and carefully reviewed hundreds of written applications and numerous interviews over the past few weeks. We only accept applicants who demonstrate genuine interest, exceptional passion, and who we believe will bring a valuable perspective to the 2026 HCAIR Asia Conference (ACON26). We have selected you because we believe you embody these valuable characteristics and will make a powerful contribution at ACON26.

The theme of this year's HCAIR Asia Conference is **Rising with Resilience**. Given rapid world development coinciding with progress in artificial intelligence, healthcare, and technology, we are choosing to promote resilience and determination among leaders. We encourage you to explore how innovation and strategic clarity might guide the next chapter of our world's development.

1,500

students reached • Hack Club

29

patches merged upstream

829

contributions, 12 months

32

repositories shipped

Recall

a memory engine

ATTEST

proof before money

Support Triage

cannot be talked into it

Minutes

a workspace, rebuilt

PDFChat

cites, or refuses

Recall

A memory engine for unfinished work. One keystroke brings back the tabs, files and chats from your last session.

0

models in the production path

THE PROBLEM

Someone interrupts you at 3pm. You sit back down and everything is still there, and none of it helps. **You did not lose files. You lost continuity.**

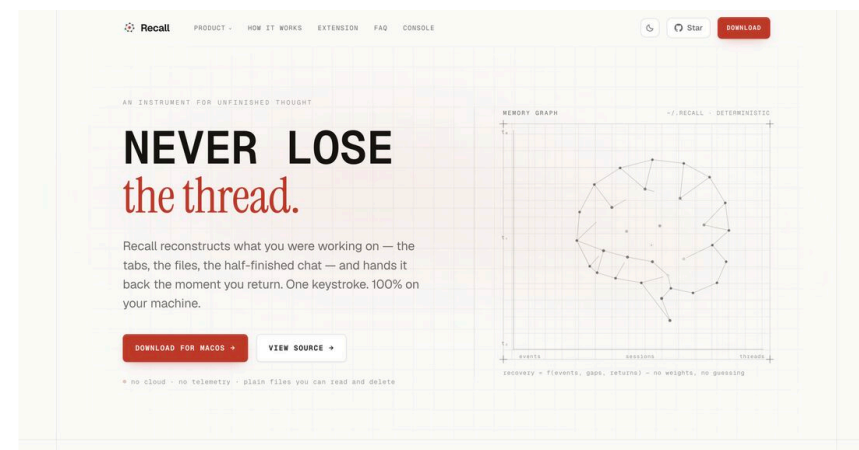
Spotlight and grep assume you remember a string. A cognitive restart asks *what was I doing*, and nothing on the machine answers that.

TECHNICAL ASPECTS

Seven layers over a local event log, fed by a Chrome extension and a file watcher. Press Ctrl + Space and the moment, the session and the topic all come back at once.

No model sits in the production path, so the same question can never return two different pasts. A 35-section smoke test enforces it.

No cloud, no account, no telemetry. Everything it knows lives in `~/.recall/` as plain JSON you can read with `cat`. Ships as a 26 MB menu-bar app that reports degraded features instead of hiding them.



YEAR	ROLE	BUILT WITH	LINK
2026	Sole author	Python · PyQt6 · FastAPI · ChromaDB · Tauri	web-seven-puce- 99.vercel.app github.com/kunaIKumar- 13/Recall-me

ATTEST

A financial control layer. It verifies what an AI proposes before anything is allowed to move money.

₹0.00

moved against the wrong account

THE PROBLEM

A payment gateway sends one lump sum covering many orders, and **more than one set of orders can add up to the same credit exactly.**

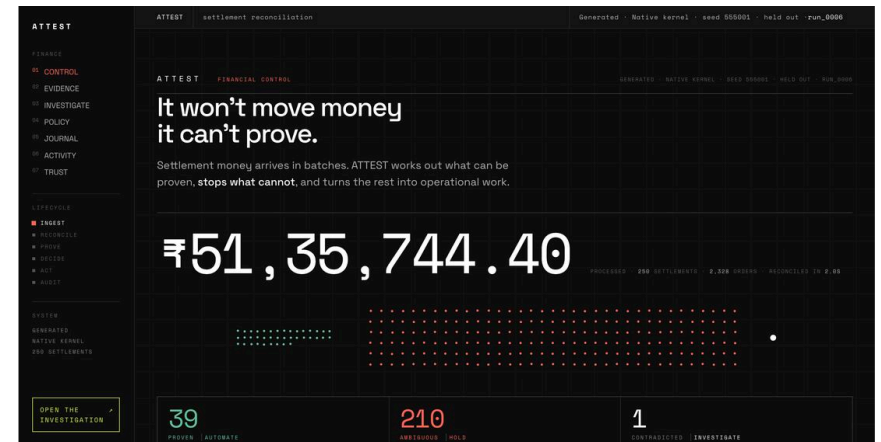
Those sets discharge *different customers*. Guessing marks the wrong customer as paid while the books still balance, so nobody finds out until it is expensive.

TECHNICAL ASPECTS

So ATTEST refuses. Across 500 settlements it had never seen, **82.4% were correctly refused**. The refusal is the product, not a failure rate.

Four stages, and the separation is the design. An **advisor** proposes which records to look at and is **never given an amount**. A **verifier** sharing no code with it re-derives every explanation. A **policy** automates only where measured error costs less than a human check.

Integer paise end to end, so no float ever touches money. The published benchmark names a competitor that beats mine on precision, because a comparison curated in your own favour proves nothing.



YEAR	ROLE	BUILT WITH	LINK
2026	Sole author	Python · Rust solver kernel	github.com/kunalKumar-13/attest

Support Triage

A support agent where code owns the control flow and the model is a constrained analyst, not the decision-maker.

132/132

evaluation tests passing

THE PROBLEM

An agent that can be talked into a destructive action is worse than no agent. The usual design hands the model the wheel and hopes the prompt holds.

This one does not. Its worst possible behaviour is *escalate to a human*.

TECHNICAL ASPECTS

The LLM is invoked exactly once per ticket, inside an unforgeable sentinel block, and returns typed JSON that a policy validator adjudicates. It cannot decide escalation, execute tools, or see raw PII.

Untrusted ticket text never leaves its frame. Defends prompt injection in seven languages plus zero-width, homoglyph and base64 obfuscation. When chunks contradict each other, **confidence drops and the validator escalates** rather than picking a side.

Two runs on the same input produce a byte-identical file.

```
$ triage --ticket 4417

  ingest    ticket text sealed in <<<TICKET>>>
 retrieve   4 chunks · 2 disagree numerically
 analyse    LLM called once, inside sentinel block
 policy     consensus broken → confidence 0.41

DECISION   ESCALATE TO HUMAN

$ sha256sum run_a.csv run_b.csv
2f9c...a71b  run_a.csv
2f9c...a71b  run_b.csv
```

YEAR	ROLE	BUILT WITH	LINK
2026	Sole author	Python · LlamaIndex · BM25 · TF-IDF	github.com/kunalKumar-13/ deterministic-support-triage

Minutes

A meeting workspace with interactive transcripts, AI notes and workspace-wide search, rebuilt from the product outward.

2

commands to a populated product

THE PROBLEM

“Clone this product” is the easiest brief in the world to do badly. You rebuild the hero section and ship something that collapses when anyone clicks.

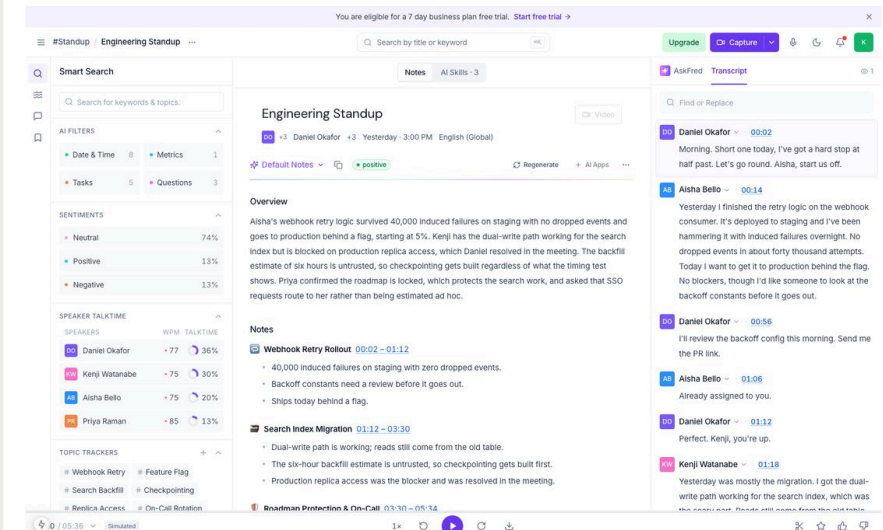
But a meeting product is not its landing page. **Everything valuable happens after the meeting ends.**

TECHNICAL ASPECTS

So I set a bar I could not fake: everything except the speech-to-text actually works, and the README says so on the first screen, before a reviewer can discover it.

Next.js 15 over FastAPI, with **SQLite FTS5** as a real full-text index rather than a LIKE query in a costume. Auth is enforced at the edge, so a signed-out visitor never receives app UI at all.

The decision that mattered: I stopped matching screens and **extracted the token system instead**, after which twenty-plus screens came out coherent untouched.



YEAR	ROLE	BUILT WITH	LINK
2026	Sole author · assignment brief	Next.js 15 · React 19 · TypeScript · FastAPI	github.com/kunalKumar-13/minutes

PDFChat

Ask a document anything. Every answer cites its pages, or says plainly that the document does not support the question.

0

uncited answers

THE PROBLEM

A confident wrong answer costs more than no answer. Retrieval gets messy, and the standard failure is a model that fills the gap fluently and invisibly.

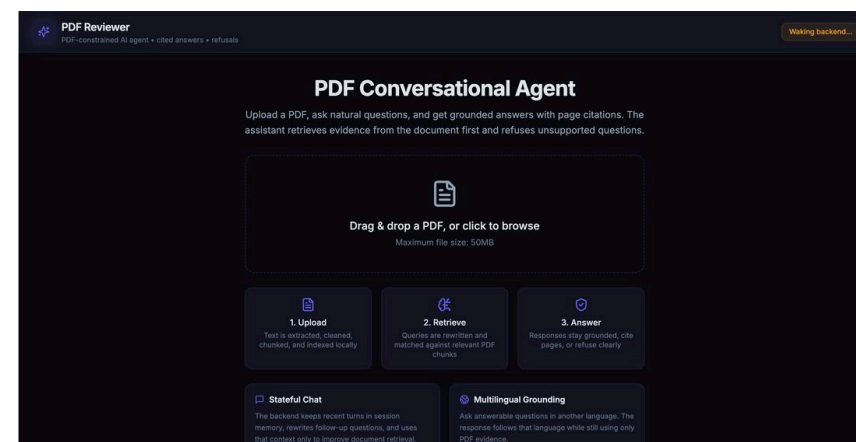
PDFChat will not.

TECHNICAL ASPECTS

Every response either **cites the pages it came from** or refuses clearly. Multi-turn chat keeps session context for retrieval only, so the document stays the sole evidence.

Extraction, chunking and indexing all run locally, and queries are rewritten against the relevant chunks before generation. It answers in your language while still treating the source as the only truth, which is exactly where grounding usually breaks quietly.

The refusal path is a first-class branch, not an error handler: if retrieval does not support an answer, that *is* the answer.



YEAR	ROLE	BUILT WITH	LINK
2026	Sole author	Python · PyMuPDF · TF-IDF · FastAPI	pdf-reviewer- ten.vercel.app github.com/kunaIKumar- 13/PDF-Reviewer

BEYOND THE FIVE PROJECTS

The rest of me

Research, rooms I did not belong in, people who owed me nothing, and 1,500 students.



1,500

students reached · 16 months as
Growth & Ops Lead, Hack Club



What I am chasing

Through to **round 2 at MATS**, and an **Activate AI Fellow**, India's first cohort, 1 of 15 from 500+.

Three projects here turned out to be one question:
when a model proposes something, what is
structurally permitted to believe it?



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Strangers grade my code

29 patches merged, 10 maintainers, four languages.
OpenStreetMap's editor, Django's own site, a Bitcoin Lightning wallet, MetaBrainz, Plone, the Fortran standard library. I had no reason to know Fortran.

I teach, and it changed how I write

50+ students through Data Structures and Algorithms. The third explanation is the one you build live, and it is the only one that works. It is why every README I ship opens by naming what the software does *not* do.

Rooms I do not belong in

Selected delegate to **HPAIR 2026 at Harvard**. Every project here began as a problem I could name in one sentence, and that **habit came from that room, not a codebase**.

Why this Academy

Everything I build lives on a screen someone is already sitting at. I have never built for the device someone holds *while something is happening to them*. An honest **"I don't know"** is worth nothing if **nobody can read it**. That is the half I am asking you to teach me.